higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 3

19 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 1.1.1 C
1.1.2 D
1.1.3 B
1.1.4 C

(4 × 1) (4)

1.2
- Completion date
- Starting date
- Available employees
- Details concerning ordering of materials
- Work permit requests
- Supply requests from authorities, e.g. Eskom or municipality
- Make provision for unforeseen circumstances (Any 3 × 1) (3)

1.3
- Provide direction
- Minimise waste
- Set standards and facilitate control
- Identify resources to achieve goals
- Decide which tasks must be done to achieve goals (Any 2 × 1) (2)

1.4
- To establish the sequence of different tasks
- To establish the time taken to complete different tasks (2)

1.5
- Eliminate hazards
- Isolate hazards
- Reduce or minimise hazards (3)

1.6
- Working with and trained by a qualified person
- Demonstrations of the operation of and repairs to newly installed equipment
- Demonstrations of how to maintain and use PPE
- Videos shown during orientation about workplace safety (Any 3 × 1) (3)

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1.6.2
- Pay attention to what employees and health and safety representatives have to say about the issues that affect them.
- Determine how injuries and diseases may possibly occur.
- Identify which hazards will cause injuries or diseases, and also how serious these injuries or diseases are likely to be.
- Make sure that employees are kept informed of the likelihood of the injuries and diseases, provide instructions on how to minimise the chances of such incidence, train employees on safety precautions and provide supervision to create awareness of health and safety issues.
- Encourage employees to report all incidents, accidents, illnesses, deaths and dangerous situations with a view to prevent them from reoccurring.
- Put measures in place to minimise stress in the workplace.
- Do whatever it takes to make sure that the workplace is safe.
- Maintain all machinery and keep the working environment in an excellent condition. (Any 3 × 1) (3)

1.7
1.7.1 False
1.7.2 False
1.7.3 True
1.7.4 True
1.7.5 False

(5 × 1) (5)
[25]

QUESTION 2

2.1

ANALOGUE METERS	DIGITAL METERS
• Have moving parts • Have a parallax error • Take measurement by reading the scale • No auto range function • Require frequent calibration	• Have no moving parts • Have no parallax error • Measurement is given • Have auto range function • Calibration once a year

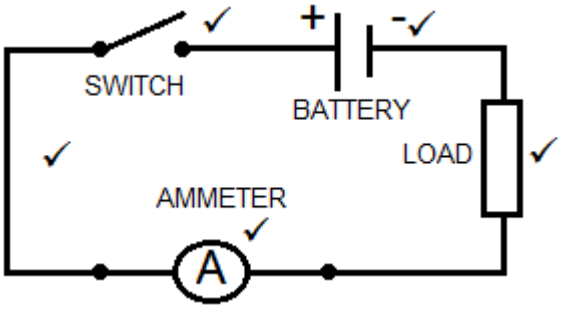
(Any 2 × 1) (2)

2.2
1.2.1 Tachometer
2.2.2 Stroboscope
2.2.3 Insulation resistance tester

(3 × 1) (3)

2.3
- Mechanical shock
- Exposure to magnetic fields
- Excessive current flow (Any 2 × 1) (2)

2.4



(ONE mark per correct symbol, ONE mark for correct circuit diagram) (5)

2.5 $\frac{Np}{Ns} = \frac{Vp}{Vs}$

$Vs = \frac{Ns \ x \ Vp}{Np}$ ✓

$Vp = \frac{400 \ x \ 60}{600}$

$Vp = \frac{24 \ 000}{600}$ ✓

$Vp = 40 \ V$ ✓

(ONE mark for the formula and TWO marks for the calculation) (3)

[15]

QUESTION 3

3.1
- Mechanical timers sequence the washing process.
- This is done using cams on a shaft in the timer.
- The cams operate switches according to the washing cycle. (3)

3.2
- Radiant heaters
- Convection heaters
- Fan heaters (3)

3.3
3.3.1 sealed
3.3.2 tube
3.3.3 break
3.3.4 temperature

(4 × 1) (4)

3.4
3.4.1 Produces very high frequencies needed for cooking
3.4.2 To keep food at serving temperature
3.4.3 To remove moisture from clothing

(3 × 1) (3)

3.5
- A dedicated circuit shall be provided for cooking appliances that are rated at more than 16 A.
- One circuit shall not supply more than one permanently connected cooking appliance unless the appliances are in the same room.
- A cooking appliance circuit may also supply one socket outlet if the rating of the socket outlet does not exceed 16 A. (Any 2 × 1) (2)

[15]

QUESTION 4

4.1
- Do not touch live conductors.
- Be more cautious when working on appliances that use water.
- Working at a customer's premises is risky, so be extra careful.
- Avoid having children or animals in your work area.
- Socket outlets not on earth leakage are dangerous.
- Insulated tools must be used. (Any 5 × 1) (5)

4.2
- Use a small screwdriver to remove the brush holder.
- Do a visual inspection of the brushes for wear.
- Replace brushes if they are worn or out of specification.
- Sometimes the brush holder becomes sticky because of accumulated dust and other materials. In such cases, clean the brush holder with a cotton bud. (4)

4.3
4.3.1 C
4.3.2 A
4.3.3 B
4.3.4 E
(4 × 1) (4)

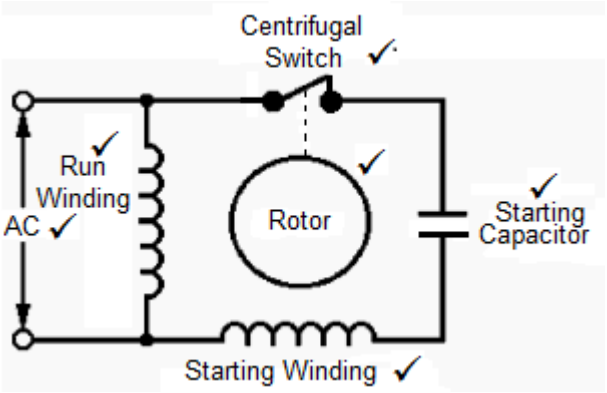
4.4
4.4.1 C
4.4.2 A
(2 × 1) (2)

[15]

QUESTION 5

5.1 Squirrel cage rotor (1)

5.2



(6)

5.3
- Coil
- Armature
- Armature spring
- Coil core
- Coil terminal
- Common terminal

(Any 2 × 1) (2)

5.4 Totally enclosed✓ with seals at each shaft end✓ to prevent water from entering the motor✓ (3)

5.5 5.5.1 Direct on line (1)

5.1.2
- Start button – green
- Stop button – red

(2)
[15]

QUESTION 6

6.1 The SMAW process involves using heat✓ from an electrical arc between the electrode and the workpiece for joining metal.✓ (2)

6.2 6.2.1 The power in a welding circuit

6.2.2 The cables that carry or supply the current to and from the workpiece

6.2.3 Used to grip the electrode

(3 × 1) (3)

6.3
- Fumes and gases
- Electric shock
- Electrocution
- Ultraviolet
- Heat radiation
- Noise
- Fire (Any 4 × 1) (4)

6.4
- Butt joint
- Corner joint
- Edge joint
- Lap joint
- T-joint (Any 4 × 1) (4)

6.5
- Head protection
- Face and eye protection
- Ear protection
- Respiratory protection
- Hand protection
- Footwear
- Protective clothing
- Welding mats (2 × 1) (2)

[15]

TOTAL: 100

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